

Lab: Thought Experiment — Action, Exploration, and Free Will

Objective

Investigate the philosophical implications of treating action as inference through thought experiments on exploration, skilled performance, and the problem of free will.

Part 1: The Rattle Experiment

Goal: Analyze epistemic action in the simplest case.

An infant shakes a rattle for the first time. The sound surprises her. She shakes it again — and again — each time with slightly different force and angle.

- Is the infant "playing" or "experimenting"? Does the distinction matter?
- How does Active Inference explain this behavior (hint: epistemic action to reduce expected ambiguity about the rattle's causal properties)?
- At what point does the infant stop exploring and move to a different toy? What does this transition reveal about the EFE balance between epistemic and pragmatic value?

Write your response here

Part 2: The Expert's Flow

Goal: Connect the phenomenology of skilled action to the Active Inference framework.

Imagine a jazz musician improvising a solo. She does not consciously plan each note; the music "flows" through her. Dreyfus calls this "absorbed coping."

- How does Active Inference account for the transition from effortful, deliberate play (the student) to effortless, absorbed play (the expert)?
- What changes in the generative model? (Hint: precision, model complexity, habitual policies.)
- Is the improvising musician "free" in her playing? More free or less free than the deliberating student?

Write your response here

Part 3: The Free Will Dilemma

Goal: Engage directly with the philosophical problem of free will.

Active Inference says that actions are determined by prediction error minimization within a generative model. Counterargument: "If my actions are the output of a deterministic computational process, I am no more free than a thermostat."

- State the strongest version of this objection.
- Present the Active Inference response (hierarchical generative models, counterfactual reasoning, Frankfurt-style higher-order volition).
- Evaluate: Does this response genuinely preserve free will, or does it merely redescribe determinism in flattering language?

Write your response here

Part 4: Moral Responsibility Under Active Inference

Goal: Explore the ethical implications of action as inference.

If a person with an atypical generative model (due to psychopathology, developmental difference, or extreme environment) performs harmful actions, are they morally responsible?

- How does Active Inference's account of action bear on the legal concept of diminished responsibility?
- Is "my generative model made me do it" a valid excuse? Why or why not?
- How does this connect to philosophical discussions of moral luck (Nagel, Williams)?

Write your response here

Part 5: Synthesis and Position Defense

Goal: Articulate your philosophical view on action and free will.

Write a 200-word statement on one of:

- "Active Inference provides the best available compatibilist account of free will."
- "Action as inference eliminates genuine agency — we are sophisticated prediction machines, nothing more."
- "The distinction between epistemic and pragmatic action recaptures the Aristotelian distinction between *theoria* and *praxis*."

Share with a peer and defend your position against their strongest objection.

Write your response here

Lab Summary

Part	Skill Developed	Key Philosophical Move
1	Developmental analysis	Interpreting infant behavior through the epistemic/ pragmatic value lens
2	Phenomenological analysis	Connecting absorbed expertise to generative model dynamics
3	Dialectical reasoning	Engaging with free will through objection and response
4	Applied ethics	Connecting action as inference to moral responsibility
5	Position defense	Articulating and defending a philosophical position on action and freedom